Time Series

Spring Semester 2026, EPFL
École Polytechnique Fédérale de Lausanne
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Practice Exam — Week 8: Spectral Estimation — Periodogram & Tapering

Duration: 60 minutes · No calculator · Answer on paper

Name: ______________________ Surname: ______________________

SCIPER (Student Card Number): ______________________

Exercise	**Points**
1	
2	
3	
4	
Total	

Justify every answer. Throughout, ε_t is a mean-zero white noise of variance σ^2.

Problem 1 (5 points)

We observe $X_1, \ldots, X_n$ ($T = \{1, \ldots, n\}$, sampling interval $\Delta = 1$). The *periodogram* is the Fourier transform of the biased sample ACVS,

$$\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau \in \mathbb{Z}} \hat{\gamma}_\tau \, e^{-2\pi i \tau f}, \qquad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} \big(X_t - \bar{X}\big)\big(X_{t+|\tau|} - \bar{X}\big)$$

for $|\tau| \leq n-1$ (and 0 otherwise). Write $Y_t = X_t - \bar{X}$ throughout.

(a) Show that

$$\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2, \qquad J(f) = \sqrt{\tfrac{1}{n}} \sum_{t \in T} Y_t \, e^{-2\pi i t f}.$$

Hint: re-index the double sum over (t, τ) as a free double sum over $(s, t) \in T \times T$. (2.5 pts)

(b) Deduce two structural properties from the representation in (a): that $\hat{\mathcal{S}}^{(p)}(f) \geq 0$ for every f, and that $\hat{\mathcal{S}}^{(p)}$ is even, $\hat{\mathcal{S}}^{(p)}(-f) = \hat{\mathcal{S}}^{(p)}(f)$. (1.5 pts)

(c) Show that $\hat{\mathcal{S}}^{(p)}(0) = 0$. (The mean removal makes the periodogram blind to the zero frequency.) (1 pt)

Problem 2 (5 points)

This problem develops the expectation (bias) of the periodogram. Assume $\{X_t\}$ is zero-mean stationary with ACVS $\{\gamma_\tau\} \in \ell^1$ and SDF $\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i\tau f}$, and replace $\bar{X}$ by the true mean $\mu = 0$, so that $Y_t = X_t$.

(a) Starting from $\hat{\mathcal{S}}^{(p)}(f) = \frac{1}{n}\sum_{t=1}^{n}\sum_{s=1}^{n} Y_t Y_s\, e^{-2\pi i f(t-s)}$, take expectations and show that

$$\mathbb{E}\big[\hat{\mathcal{S}}^{(p)}(f)\big] = \sum_{\tau=-(n-1)}^{n-1} w_\tau\, \gamma_\tau\, e^{-2\pi i\tau f}, \qquad w_\tau = 1 - \frac{|\tau|}{n}.$$

(2 pts)

(b) The triangular weight is $w_\tau = \frac{1}{n}\sum_t d_t\, d_{t+|\tau|}$, the autocorrelation of the rectangular data window $d_t = \mathbf{1}_T(t)$. Using the discrete convolution theorem, show that the Fourier transform of $\{w_\tau\}$ is the **Fejér kernel**

$$\mathcal{F}_n(f) = \frac{1}{n}\,|D(f)|^2, \qquad D(f) = \sum_{t=1}^{n} e^{-2\pi i t f},$$

and conclude that $\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f')\,\mathcal{F}_n(f-f')\,df'$, i.e. the expected periodogram is the true SDF *blurred* by $\mathcal{F}_n$. (2 pts)

(c) Show that $\mathcal{F}_n$ is a genuine averaging kernel by proving the normalisation $\int_{-1/2}^{1/2} \mathcal{F}_n(f)\,df = 1$. Then state in one sentence why the periodogram is nevertheless *asymptotically* unbiased as $n \to \infty$. (1 pt)

Problem 3 (5 points)

This problem treats *tapering*, the cure for periodogram bias. A **data taper** is a sequence $\{h_t\}_{t\in T}$ (extended by $h_t = 0$ off T) with $\|h\|_2^2 = \sum_t h_t^2 = 1$. With μ removed ($Y_t = X_t - \mu$), the tapered periodogram is

$$\hat{\mathcal{S}}_h^{(p)}(f) = |J_h(f)|^2\,, \qquad J_h(f) = \sum_{t\in T} h_t\, Y_t\, e^{-2\pi i t f},$$

and $H(f) = \sum_{t\in\mathbb{Z}} h_t e^{-2\pi i t f}$ is the Fourier transform of the taper.

(a) For zero-mean stationary $\{X_t\}$, show that

$$\mathbb{E}\big[\hat{\mathcal{S}}_h^{(p)}(f)\big] = \sum_{\tau\in\mathbb{Z}} w_\tau\, \gamma_\tau\, e^{-2\pi i \tau f}, \qquad w_\tau = \sum_{t\in\mathbb{Z}} h_t\, h_{t+\tau},$$

and, using the convolution theorem on $w = \tilde{h} * \bar{h}$ (with $\tilde{h}_t = h_{-t}$, $\bar{h}_t = h_t$), deduce the kernel form

$$\mathbb{E}\big[\hat{\mathcal{S}}_h^{(p)}(f)\big] = \int_{-1/2}^{1/2} \mathcal{S}(f')\, \big|H(f-f')\big|^2\, df'.$$

(2.5 pts)

(b) *(Unbiasedness for white noise.)* Suppose $\{X_t\} = \{\varepsilon_t\}$ is white noise, so $\mathcal{S} \equiv \sigma^2$. Using Parseval, $\int_{-1/2}^{1/2} |H(f)|^2\, df = \sum_t h_t^2$, prove that

$$\mathbb{E}\big[\hat{\mathcal{S}}_h^{(p)}(f)\big] = \sigma^2\, \|h\|_2^2\,,$$

and conclude that $\hat{\mathcal{S}}_h^{(p)}(f)$ is *exactly* unbiased for every n and every f precisely because $\|h\|_2^2 = 1$. (1.5 pts)

(c) *(Multitapering.)* Given K orthonormal tapers $\{h_{t,k}\}$ with $\sum_t h_{t,k} h_{t,k'} = \delta_{k,k'}$, the multitaper estimator is $\hat{\mathcal{S}}^{(mt)}(f) = \frac{1}{K}\sum_{k=1}^{K} \hat{\mathcal{S}}_{h_k}^{(p)}(f)$. Assuming the K individual tapered periodograms are asymptotically uncorrelated, each with variance $\to \mathcal{S}(f)^2$, derive $\mathrm{Var}\big(\hat{\mathcal{S}}^{(mt)}(f)\big) \to \mathcal{S}(f)^2/K$, and explain in one sentence why this (unlike the raw periodogram) restores consistency. (1 pt)

Problem 4 (5 points)

(Revision of Week 7: spectral density of an MA process; the valid-ACVS test.) Throughout, frequencies are ordinary ($\Delta = 1$) and ε_t is white noise of variance σ^2.

(a) Let $\{X_t\}$ be the MA(2) process $X_t = \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2}$. Compute its ACVS $\gamma_0, \gamma_1, \gamma_2$ (and note $\gamma_\tau = 0$ for $|\tau| \geq 3$), and show that its SDF equals

$$\mathcal{S}(f) = \sigma^2 \left|1 + \theta_1 e^{-2\pi i f} + \theta_2 e^{-4\pi i f}\right|^2.$$

(2 pts)

(b) Evaluate $\mathcal{S}(f)$ in the fully real cosine form $\mathcal{S}(f) = \gamma_0 + 2\gamma_1\cos(2\pi f) + 2\gamma_2\cos(4\pi f)$ for the values $\theta_1 = 1$, $\theta_2 = \frac{1}{4}$, $\sigma^2 = 1$. Give $\mathcal{S}(0)$ and $\mathcal{S}(\frac{1}{2})$, and verify non-negativity is consistent with the process being a genuine MA. (1.5 pts)

(c) *(Valid-ACVS test.)* Consider the candidate sequence $\gamma_0 = 1$, $\gamma_{\pm 1} = a$, and $\gamma_\tau = 0$ for $|\tau| \geq 2$ (with $a \in \mathbb{R}$). Compute the associated $\mathcal{S}(f) = 1 + 2a\cos(2\pi f)$, and determine the exact range of a for which $\{\gamma_\tau\}$ is a valid ACVS of some stationary process. (1.5 pts)